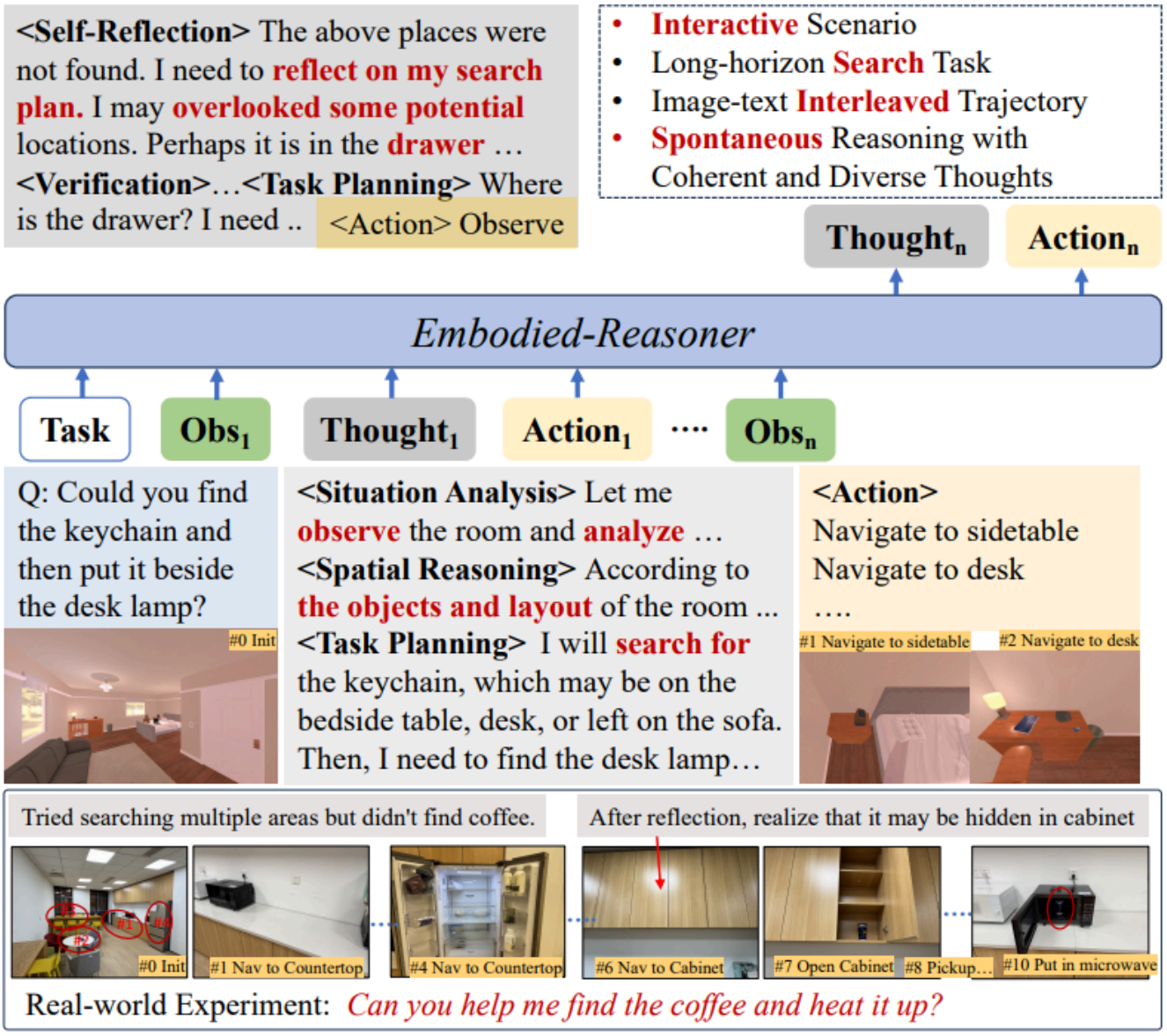
Embodied-Reasoner: Synergizing Visual Search, Reasoning, and Action for Embodied Interactive Tasks

> #EmbodiedAI #SFT #Dataset



- What problem does this framework solve?
  - LLMs with complicated thinking ability are able to solve hard tasks. However, it still remains unexplored in the embodied field.
  - In such environments, the models should not only handle the context with both images and texts but also need to be able to understand the scene, physical laws, and the space.
  - Current frameworks will have hallucinations and goal forgetting when they handle long-horizon complex embodied tasks.
- What does this framework do?
  - Dataset/Data Engine

        - Based on AI2-Thor simulator, multiple templates, including search, navigation, and transportation, are created.
        - Based on the metadata of the scene, Affiliation Graphs are created (e.g., `key in box in livingroom`). Based on these affiliation graphs, the system can find the shortest way to achieve the goal (e.g., find the key).
        - To simulate the actual searching, some additional searching processes are added.
        - Interleaving Thought: Between each observation and action, 5 thinking processes are added
            1. Situation Analysis
            2. Task Planning
            3. Spatial Reasoning
            4. Self-reflection
            5. Double Verification
    - 3-step SFT: Finetuning the Qwen2-VL-7B model.
        - Learn to Interact: Use the data engine to produce some trajectories with basic search with simple actions to finetune the base model. After that, the model will be able to output actions but still cannot search/explore.
        - Learn to Search: Let the finetuned model from step 1 explore freely the environment at a high temperature. Filter out the failed trajectories with the data engine as a reward model. Use the successful trajectories to finetune the model. Afterwards, the model can explore and plan.
        - Learn to Self-reflect: Rerun the exploration and collect the trajectories based on the model from last step. With the successful trajectories, inject noise and error signals, and then let the model retry the task; for the failed ones, let the model generate reflection at the place where the 1st mistake happens and retry the task. **The loss will be calculated for the reflection and correct part of the output.** After this, the model will be robust and have a strong ability of self-reflection.
- Experiment Results
    - Significantly outperforms existing models: Across 809 new test tasks on the AI2-THOR simulator, Embodied-Reasoner achieved a success rate that significantly surpassed that of state-of-the-art visual reasoning models. For example, it outperformed OpenAI o1, o3-mini, and Claude-3.7 by 9%, 24%, and 13%, respectively.
    - Clear Advantages in Long-Sequence Complex Tasks: In the most complex composite tasks, its performance surpassed that of the runner-up (GPT-4o) by a staggering 39.9%. Experiments revealed that as task complexity increases, the model automatically increases the number of thinking tokens (expanding by up to nearly 5 times) to handle complex planning.
    - Significantly reduces redundant searches: Unlike baseline models that often search the same cabinet repeatedly like headless flies, Embodied-Reasoner demonstrates exceptional temporal reasoning and memory capabilities, reducing the rate of redundant exploration (RER) by approximately 50%.
    - Real-world generalization capabilities: In 30 search tasks conducted in real rooms (the physical world), using real-time image input from cameras worn by humans, the model achieved a 56.7% success rate, outperforming OpenAI's o1 (50%) and o3-mini (43.4%).

- Work Cited
    - Paper
    - Chat History
    - NotebookLM